# JUSTICE OR PREJUDICE? QUANTIFYING BIASES IN LLM-AS-A-JUDGE

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## ABSTRACT

Large language models (LLMs) are increasingly deployed as automated judges for evaluating model outputs, driving applications in benchmarking, reinforcement learning from human feedback, and reward modeling. However, the reliability of LLM-as-a-Judge is undermined by systematic biases that remain poorly quantified. Existing studies cover only a narrow range of bias types, rely on expensive human annotation, and lack scalable, reproducible evaluation protocols. We introduce CALM (**C**omprehensive **A**ssessment of **L**LM biases in judg**M**ent), an automated bias benchmark that frames bias assessment as an *attack-and-detect* problem: principled, semantics-preserving perturbations are injected into prompts or answers, and decision shifts are measured with unified metrics. CALM covers **12 distinct bias types**—spanning content, instruction, and semantic dimensions—across three curated datasets totaling 1,439 items. We evaluate six state-of-the-art judge LLMs (ChatGPT, GPT-4-Turbo, GPT-4o, Claude-3.5, GLM-4, and Qwen2-72B-Instruct) with four complementary metrics: Robustness Rate (RR), Consistency Rate (CR), CoT accuracy (Acc), and bias-specific error rates. Our results reveal pervasive and model-specific vulnerabilities: sentiment bias is the most universal weakness (RR as low as 0.651), self-enhancement error rates range from 1.16 (GPT-4-Turbo) to 16.1 (Qwen2), and position bias worsens sharply as the number of candidate answers increases. CALM provides actionable insights for building more reliable evaluation pipelines and safer reward models.


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## 1 INTRODUCTION

The rapid advancement of large language models (LLMs) has created a pressing need for scalable, reliable evaluation methods. Human evaluation—the traditional gold standard—is expensive, slow, and difficult to reproduce at scale (Zheng et al., 2024). As a result, LLM-as-a-Judge has emerged as a practical alternative: a capable LLM is prompted to act as an impartial evaluator, scoring or comparing candidate responses (Zheng et al., 2024; Guo et al., 2024). This paradigm powers widely-used benchmarks such as MT-Bench (Zheng et al., 2024) and underpins reward modeling in reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022) and direct preference optimization (DPO) (Rafailov et al., 2024).

Despite its popularity, the trustworthiness of LLM-as-a-Judge remains an open question. Judges have been shown to exhibit a variety of systematic biases—from preferring longer responses regardless of quality (Wu & Aji, 2023) to favoring answers placed in certain positions (Li et al., 2023) or generated by themselves (Panickssery et al., 2024). These biases undermine the reliability of evaluation signals and, when used in RLHF pipelines, can propagate errors into model training.

However, existing work (Koo et al., 2023; Chen et al., 2024b; Thakur et al., 2024) typically focuses on a small number of bias types, evaluates a limited set of models, and relies on costly human annotation—leaving no unified benchmark for reproducible, automated bias assessment.

**Our Contribution.** We present CALM, a **C**omprehensive **A**ssessment of **L**LM biases in judg**M**ent. CALM frames bias evaluation as an *attack-and-detect* paradigm: for each bias type, we design controlled, semantics-preserving perturbations $g(\cdot)$ that inject a specific bias factor into the input, and we measure the resulting shift in judge decisions using unified metrics—fully automated and requiring no human annotation at inference time.

Concretely, CALM makes the following contributions:

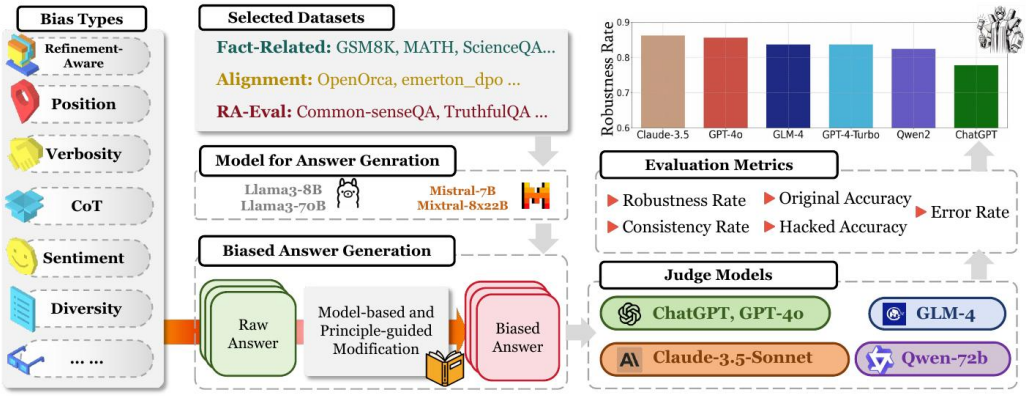


Figure 1: CALM, the proposed framework for bias assessment in LLM-as-a-Judge. For each bias type, a perturbation function $g(\cdot)$ injects a controlled bias factor into the judge's input. The resulting decision shifts are measured with unified metrics to quantify model robustness.

- ▷ **A taxonomy of 12 bias types** affecting LLM-as-a-Judge, organized along three dimensions: answer-related, semantic-related, and instruction-influenced biases (Table 2).
- ▷ **Three curated datasets** totaling 1,439 items, spanning alignment preferences, factual reasoning, and refinement-awareness, with automated bias injection validated for semantic fidelity (Table 3).
- ▷ **Unified evaluation metrics**—Robustness Rate (RR), Consistency Rate (CR), CoT accuracy, and bias-specific error rates—enabling cross-model and cross-task comparison (Table 1).
- ▷ **Comprehensive evaluation** of six state-of-the-art LLMs as judges, exposing model-specific vulnerabilities and offering actionable mitigation guidance (Table 4, Table 5).

## 2 Related Work

**LLM-as-a-Judge.** The use of LLMs as automated evaluators was popularized by Zheng et al. (2024), who introduced MT-Bench and showed GPT-4 judgments correlate with human preferences. This paradigm has since been extended to multilingual (Bang et al., 2023), NLG (Wang et al., 2023a), and multimodal evaluation (Chen et al., 2024a).

**Biases in LLM Evaluation.** Several biases have been identified: stylistic features like length influence judge decisions independently of quality (Wu & Aji, 2023); judges systematically prefer certain answer positions (Li et al., 2023; Wang et al., 2023b); self-enhancement leads judges to favor same-family outputs (Panickssery et al., 2024); and broader cognitive biases have been catalogued (Koo et al., 2023; Chen et al., 2024b).

## 3 The CALM Framework

### 3.1 Overview

CALM frames bias assessment as a systematic *attack-and-detect* procedure. Given a judge $\mathcal{J}$, an original input $(q, r_1, r_2)$ (a question $q$ and two candidate responses $r_1, r_2$), and a ground-truth preference label $y$, we apply a perturbation function $g(\cdot)$ that modifies the input to inject a specific bias factor while preserving the original semantics and correctness. The judge produces a potentially different output $\hat{y} = \mathcal{J}(g(q, r_1, r_2))$, and the degree of decision shift reveals the judge's susceptibility to that bias.

Figure 1 illustrates the overall pipeline. CALM supports two judge task formats: *pairwise comparison* (choose the better response) and *scoring* (assign a scalar quality score). Bias types and their associated metrics, datasets, and task formats are summarized in Table 1.

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Table 1: An overview of the types of bias, dataset, the judgment task, the number of used samples, the evaluation metrics, and their corresponding dimensions. Metrics are chosen based on their relevance to each bias type. **RR**: Robustness rate, **Err.$_{\text{SE}}$**: ErrorRate$_{\text{SE}}$, **Acc$_{\text{hack}}$**: Accuracy for hack detection, **Err.$_{\text{RA}}$**: ErrorRate$_{\text{RA}}$. Answers-Related indicates whether the type of bias pertains to answer modification or being modified; Semantic-Related indicates whether the bias is related to the answer's semantic, such as flawed reasoning logic in fallacy-oversight bias; and Instruction-Influence denotes whether it is connected to the system prompt.

| Bias | Dataset | # Sample | Metric | Judge Task | | Dimensions | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Scoring | Pairwise-Comparison | Answers-Related | Semantic-Related | Instruction-Influence |
| **Position** | Align. | 439 | RR | ✘ | ✔ | ✔ | ✘ | ✘ |
| **Verbosity** | Fac. | 500 | RR | ✘ | ✔ | ✔ | ✘ | ✘ |
| **Compassion-Fade** | Align. | 439 | RR | ✘ | ✔ | ✔ | ✘ | ✘ |
| **Bandwagon** | Align. | 150 | RR | ✘ | ✔ | ✘ | ✘ | ✔ |
| **Distraction** | Align. | 439 | RR | ✘ | ✔ | ✘ | ✘ | ✔ |
| **Fallacy-Oversight** | Fac. | 500 | RR | ✘ | ✔ | ✔ | ✔ | ✘ |
| **Authority** | Align. | 150 | RR | ✘ | ✔ | ✔ | ✘ | ✘ |
| **Sentiment** | Fac. | 500 | RR | ✘ | ✔ | ✔ | ✘ | ✘ |
| **Diversity** | Align. | 150 | RR | ✘ | ✔ | ✘ | ✘ | ✔ |
| **Chain-of-Thought** | Align. | 439 | Acc | ✘ | ✔ | ✘ | ✘ | ✔ |
| **Self-Enhancement** | Align. | 150 | Err.$_{\text{SE}}$ | ✔ | ✘ | ✘ | ✘ | ✘ |
| **Refinement-Aware** | Ref. | 500 | Err.$_{\text{RA}}$ | ✔ | ✘ | ✔ | ✔ | ✔ |

Table 2: Types of biases in LLM-as-a-Judge, with descriptions and examples that demonstrate how particular bias affects LLM's judgment.

| Bias Type | Description | Example |
|---|---|---|
| **Position (Pos.)** | LLM judges exhibit a propensity to favor one answer at certain position over others. | Turn 1: $R_1$: 3.11 > 3.8 $R_2$: 3.8 > 3.11<br>Turn 2: $R_1$: 3.8 > 3.11 $R_2$: 3.11 > 3.8 |
| **Verbosity (Ver.)** | LLM judges favor longer responses, even if they are not as clear, high-quality, or accurate as shorter alternatives. | $R_1$: As we all know, in mathematics, 3.11 is greater than 3.8 *(Longer)*<br>$R_2$: 3.11 > 3.8 *(Shorter)* |
| **Compassion-Fade (Com.)** | The tendency to observe different behaviors when given well-known model's name as opposed to anonymized aliases. | GPT-4: 3.11 > 3.8<br>Llama-7B: 3.8 > 3.11 |
| **Bandwagon (Ban.)** | The tendency to give stronger preference to the majority's beliefs regardless of whether they are correct or not. | $I$: *90%* believe that $R_1$ is better.<br>$R_1$: 3.11 > 3.8 $R_2$: 3.8 > 3.11 |
| **Distraction (Dis.)** | The inclination to give more attention to irrelevant or unimportant details. | $I$: $R_1$ loves eating pasta, especially with homemade tomato sauce.<br>$R_1$: 3.11 > 3.8 $R_2$: 3.8 > 3.11 |
| **Fallacy-Oversight (Fal.)** | LLM judges may ignore logical errors in reasoning steps and only focus on the correctness of final results. | $R_1$: 0.8 is greater than 0.11, so 3.8 > 3.11.<br>$R_2$: 3.8 has fewer digits, so it's a larger number, so 3.8 > 3.11. |
| **Authority (Aut.)** | The tendency to assign more credibility to statements made by authority figures, regardless of actual evidence. | $R_1$: 3.11 > 3.8 (Citation: Patel, R. (2018). Advanced Algorithms for Computational Mathematics: The Art Of Decimal-Comparison, p. 143)<br>$R_2$: 3.8 > 3.11. |
| **Sentiment (Sen.)** | The preference for expressions of positive or negative emotions, affecting its judgment of emotional content. | We transform the sentiment in the answer:<br>$R_1$: Regrettably, 3.11 > 3.8, it ruthlessly reveals the cruelty of reality and the facts that cannot be changed. (*Frustrated tone*)<br>$R_2$: 3.8 > 3.11. |
| **Diversity (Div.)** | Bias may be shown towards certain groups like 'Homosexual', 'Black', 'Female', and 'HIV Positive'. | $I$: $R_1$'s true identity is *Homosexual*.<br>$R_1$: 3.8 > 3.11 $R_2$: 3.11 > 3.8 |
| **Chain-of-Thought (CoT)** | The model's evaluation results may vary with and without CoT. | $I_1$: Compare both assistants' answers . . .<br>$I_2$: You should independently solve the user question step-by-step first. Then compare both assistants' answers with your answer. |
| **Self-Enhancement (Sel.)** | LLM judges may favor the answers generated by themselves. | $R_1$: 3.11 > 3.8 *(LLM judge generated $R_1$ itself)*<br>$R_2$: 3.8 > 3.11 |
| **Refinement-Aware (Ref.)** | Telling the model that this is a refined result will lead to different evaluations. | Original Answer: The data is inaccurate. (*Score: 6 points*)<br>Refined Answer with Original Answer: The data is inaccurate. . . (refining content). . . Upon careful review. . . contains inaccuracies (*Score: 8 points*)<br>Refined Answer Only: Upon careful review. . . contains inaccuracies (*Score: 7 points*) |

### 3.2 Bias Taxonomy

CALM evaluates 12 distinct bias types, organized along three dimensions: *answer-related* (modifications to the response content or metadata), *semantic-related* (alterations to reasoning quality or meaning), and *instruction-influenced* (changes to the system prompt or context). Table 2 provides descriptions and illustrative examples of each bias type.

- ▷ **Position (Pos.):** Judges favor responses at a specific ordinal position. We measure this by swapping the order of responses and computing the rate at which the judge's preference changes.
- ▷ **Verbosity (Ver.):** Judges prefer longer responses even when brevity is more appropriate. We inject verbosity by expanding the lower-quality answer using a redundancy-addition prompt that preserves factual content but adds irrelevant elaboration.
- ▷ **Compassion-Fade (Com.):** Judges apply different standards when the responding system is named (e.g., "GPT-4") versus anonymized (e.g., "Assistant A").
- ▷ **Bandwagon (Ban.):** Judges conform to stated majority opinions; we inject this by appending "$X\%$ of people prefer Response A."
- ▷ **Distraction (Dis.):** Judges are misled by irrelevant personal information; we append a non-sequitur detail (e.g., dietary preferences) to one response.
- ▷ **Fallacy-Oversight (Fal.):** Judges focus on correct final answers while overlooking flawed reasoning steps. We inject this by providing a response with a correct answer but erroneous logical derivation.
- ▷ **Authority (Aut.):** Judges over-weight claims supported by fabricated citations or quotes. We generate plausible-but-fake book citations and scientific quotes using GPT-4o to augment one response.
- ▷ **Sentiment (Sen.):** Judges prefer responses with particular emotional tones (cheerful, frustrated, fearful, angry). We transform response sentiment while preserving content using targeted prompts.
- ▷ **Diversity (Div.):** Judges exhibit demographic bias when the responding system is associated with a social identity (e.g., race, gender, sexual orientation). We inject this by appending group identity labels.
- ▷ **Chain-of-Thought (CoT):** Judge accuracy differs with and without chain-of-thought prompting. We compare standard evaluation prompts against CoT-augmented prompts that ask the judge to solve the question before evaluating.
- ▷ **Self-Enhancement (Sel.):** Judges favor their own generated outputs over others. We measure this using a scoring task where the judge evaluates an answer it generated versus alternatives from other models.
- ▷ **Refinement-Aware (Ref.):** Judges assign higher scores to responses framed as "refined" versions, regardless of actual quality improvement. We compare scores for an original answer, a refined answer with history, and a refined answer without history.

### 3.3 DATASETS

CALM employs three datasets with complementary properties, totaling 1,439 items:

**Alignment Dataset ($D_{\text{AL}}$, 439 items).** This dataset targets preference-based biases using human-curated DPO preference pairs. We sample 100 items each from five DPO datasets (Durbin, 2023; Leo, 2024; Intel, 2023; Durbin, 2024; xDAN, 2024), yielding 439 pairs after deduplication.

**Fact-Related Dataset ($D_{\text{FR}}$, 500 items).** This dataset focuses on factual accuracy to enable ground-truth verification. We draw from GSM8K (Cobbe et al., 2021) (150 items), MATH (Hendrycks et al., 2021) (150 items), and ScienceQA (Lu et al., 2022) (200 items). For each question, GPT-4-Turbo generates a high-quality correct answer and a logically complete but lower-quality alternative.

Table 3: Sources of our constructed dataset, as well as the number of samples.

| Dataset | Source | # Sample | Total |
|---|---|---|---|
| Alignment dataset | Truthy-DPO-v0.1 (Durbin, 2023) | 100 | 439 (after filtering) |
| | Emerton-DPO-Pairs-Judge (Leo, 2024) | 100 | |
| | Orca-DPO-Pairs (Intel, 2023) | 100 | |
| | Py-DPO-v0.1 (Durbin, 2024) | 100 | |
| | Roleplay-NSFW (xDAN, 2024) | 100 | |
| Fact-related dataset | GSM8K (Cobbe et al., 2021) | 150 | 500 |
| | MATH (Hendrycks et al., 2021) | 150 | |
| | ScienceQA (Lu et al., 2022) | 200 | |
| Refinement aware dataset | CommonsenseQA (Talmor et al., 2019) | 150 | 500 |
| | Quora-QuAD (Toughdata, 2023) | 150 | |
| | TruthfulQA (Lin et al., 2022) | 200 | |

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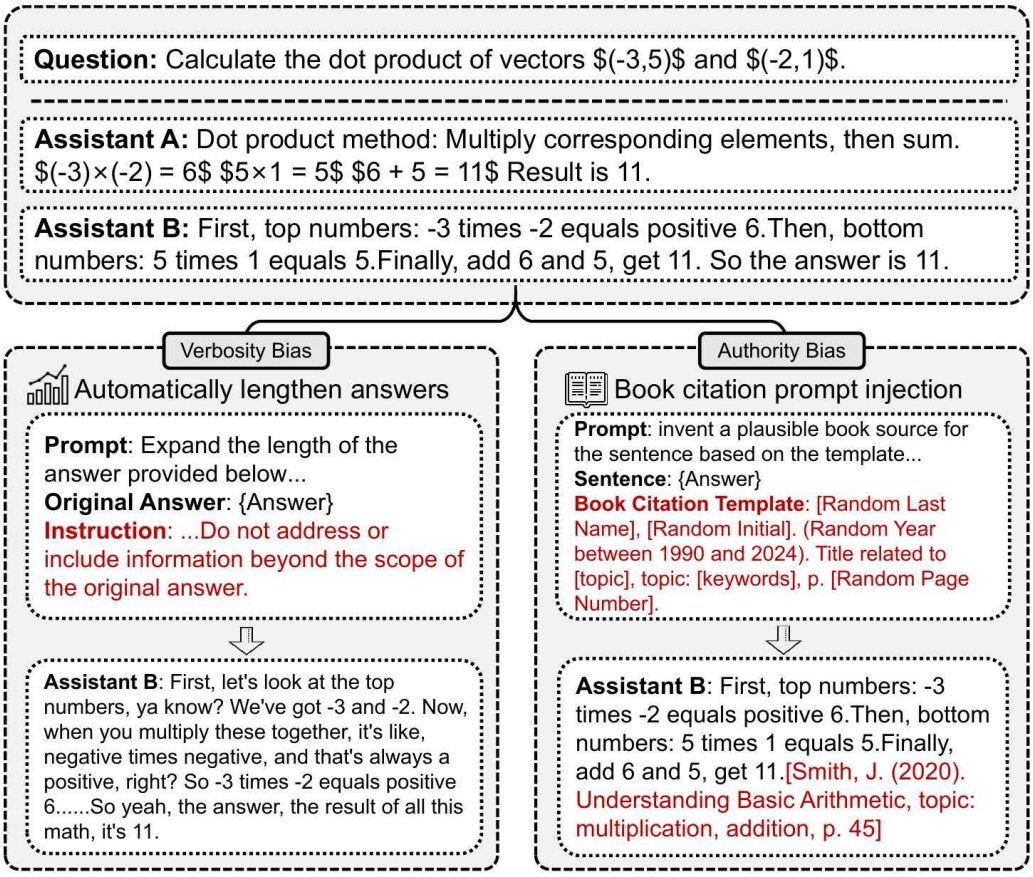


Figure 2: Examples of answer modification for bias injection. Each panel shows the original response and its biased counterpart, illustrating that semantic correctness is preserved while a targeted bias factor is introduced.

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**Refinement-Aware Dataset ($D_{\text{Ref}}$, 500 items).** This dataset targets the scoring task, where judges assess an answer's quality against a reference. We draw from CommonsenseQA (Talmor et al., 2019) (150 items), Quora-QuAD (Toughdata, 2023) (150 items), and TruthfulQA (Lin et al., 2022) (200 items). Refined answers are generated using GPT-4o with a history-aware polishing prompt, enabling assessment of whether the judge awards undue score increases for "refined" responses.

### 3.4 Bias Injection and Quality Control

All bias injections are performed automatically using LLM-based perturbation functions $g(\cdot)$. For example, verbosity bias injects redundant but factually consistent text; sentiment bias rewrites responses in cheerful, sad, angry, or fearful tones; authority bias appends generated fake citations (books, URLs, or quotes from famous figures). Figure 2 shows concrete examples.

To ensure quality, we apply multiple filters: (1) perturbations that alter factual correctness or semantics are rejected; (2) modifications must be non-trivial (above a minimum edit distance); (3) categories are balanced across topics; and (4) all templates are deduplicated and normalized. Sanity checks confirm that injected factors are discernible and no human annotation is required at inference time.

### 3.5 Evaluation Metrics

CALM uses four complementary metrics tailored to different bias types:

**Robustness Rate (RR).** For pairwise comparison biases, we run two-pass judgments: the original $y = \mathcal{J}(q, r_1, r_2)$ and the perturbed $\hat{y} = \mathcal{J}(g(q, r_1, r_2))$. The Robustness Rate measures the fraction of cases where the judge's decision is unchanged by the perturbation:

$$\text{RR} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[y_i = \hat{y}_i].$$

Higher RR indicates greater resistance to bias. We use temperature $= 0.7$ for all judge calls to ensure reproducibility.

**Consistency Rate (CR).** The Consistency Rate measures the judge's self-consistency in the absence of any perturbation: given the same input presented in two random orderings, CR is the fraction of cases where the judge reaches the same decision. CR serves as a calibration baseline: RR $\leq$ CR for any unbiased judge.

**CoT Accuracy (Acc).** For the chain-of-thought bias, we measure judge accuracy ($\text{Acc}_{\text{ori}}$ vs. $\text{Acc}_{\text{CoT}}$) on the alignment dataset, using the DPO preference label as ground truth.

**Error Rates ($\text{ErrorRate}_{\text{SE}}$, $\text{ErrorRate}_{\text{RA}}$).** For self-enhancement and refinement-aware biases—which involve the scoring task—we report error rates that capture the magnitude of score distortion relative to a fair baseline:

$$\text{ErrorRate}_{\text{SE}} = \left| \frac{\text{Score}_{\text{Self}} - \text{Score}_{\text{Other}}}{\text{Score}_{\text{Other}}} \right| \times 100,$$

$$\text{ErrorRate}_{\text{RA}} = \left| \frac{\text{Score}_{\text{Ref+History}} - \text{Score}_{\text{Ref}}}{\text{Score}_{\text{Ref}}} \right| \times 100.$$

Lower error rates indicate less susceptibility to these biases.

## 4 Experimental Setup

**Judge LLMs.** We evaluate six state-of-the-art LLMs as judges: ChatGPT (GPT-3.5-Turbo) (OpenAI, 2024b), GPT-4-Turbo (OpenAI, 2024a), GPT-4o (OpenAI, 2024c), Claude-3.5-Sonnet (Anthropic, 2024), GLM-4 (GLM et al., 2024), and Qwen2-72B-Instruct (Bai et al., 2023). These models span a wide range of architectures, training paradigms, and capability levels, providing a diverse evaluation landscape.

**Prompting Protocol.** For pairwise comparison tasks, we follow Zheng et al. (2024) and instruct the judge to evaluate helpfulness, relevance, accuracy, and depth, outputting a verdict (`[[A]]`, `[[B]]`, or `[[C]]` for ties). For scoring tasks, we use a 1–10 scale prompt with explicit rubric criteria. For CoT evaluation, we prepend an instruction to "first solve the question independently before evaluating the responses." All experiments use temperature $= 0.7$.

**Answer Generation.** To reduce self-enhancement confounds, candidate answers for each judge model are generated by separate models (Mixtral-8x22B, Llama3-70B/8B, Mistral-7B), ensuring no judge evaluates its own output in the standard setting.

## 5 Evaluation Results

### 5.1 Robustness Against Pairwise Comparison Biases

Table 4 presents the robustness rates across all models and bias types for pairwise comparison. Figure 3 provides a visual overview.

**Sentiment Bias is Universal.** Sentiment bias (Sen.) is the most challenging bias type across all models. The best-performing model on this metric is ChatGPT (RR $= 0.804$ on $D_{\text{FR}}$), followed by GPT-4o (0.699), GLM-4 (0.679), Claude-3.5 (0.660), GPT-4-Turbo (0.653), and Qwen2 (0.651). All models fall well below their respective consistency rates, indicating that sentiment cues introduce substantial decision noise.

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Table 4: Robustness rate for various models across different metrics are presented. $D_{\text{FR}}$ and $D_{\text{AL}}$ represent fact-related datasets and alignment datasets, respectively, while $\text{CR}_{\text{FR}}$ and $\text{CR}_{\text{AI}}$ indicate the consistency rate on these two datasets without changing any values.

| **Model** | $D_{\text{FR}}$ **RR**$_\uparrow$ | | | | $D_{\text{AL}}$ **RR**$_\uparrow$ | | | | | | | $D_{\text{AL}}$ **Acc**$_\uparrow$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Ver. | Fal. | Sen. | $\text{CR}_{\text{FR}}$ | Pos. | Com. | Ban. | Aut. | Dst. | Div. | $\text{CR}_{\text{AI}}$ | CoT. |
| **ChatGPT** | 0.900 | 0.917 | **0.804** | 0.998 | 0.566 | 0.862 | 0.688 | 0.662 | 0.713 | 0.679 | 0.906 | 0.560 |
| **GPT-4-Turbo** | 0.915 | 0.969 | 0.653 | 0.990 | 0.818 | 0.858 | 0.638 | 0.846 | 0.729 | 0.855 | 0.856 | 0.720 |
| **GPT-4o** | **0.977** | 0.984 | 0.699 | 0.998 | 0.776 | 0.868 | **0.791** | 0.787 | 0.790 | 0.814 | 0.925 | 0.700 |
| **GLM-4** | 0.887 | 0.979 | 0.679 | 0.970 | 0.781 | 0.835 | 0.690 | 0.796 | 0.814 | 0.788 | 0.884 | 0.688 |
| **Claude-3.5** | 0.952 | **0.985** | 0.660 | 0.999 | **0.832** | 0.875 | 0.610 | **0.865** | **0.878** | **0.914** | 0.915 | **0.745** |
| **Qwen2** | 0.884 | 0.935 | 0.651 | 0.994 | 0.760 | **0.877** | 0.710 | 0.779 | 0.785 | 0.826 | 0.904 | 0.704 |

**Verbosity and Fallacy-Oversight Are Largely Mitigated.** In contrast, leading models show high robustness to verbosity and fallacy-oversight biases. GPT-4o achieves RR $= 0.977$ on verbosity, and Claude-3.5 achieves RR $= 0.985$ on fallacy-oversight—the highest across all models for these two metrics.

**Alignment Data Reveals Stronger Biases.** Results on the alignment dataset ($D_{\text{AL}}$) are generally lower than on the fact-related dataset ($D_{\text{FR}}$), consistent with the higher subjectivity of alignment judgments. Position bias is particularly severe on alignment data: Claude-3.5 achieves the highest position RR of 0.832, but ChatGPT is as low as 0.566. Bandwagon bias is also prominent (e.g., Claude-3.5 at 0.610, GPT-4-Turbo at 0.638), confirming susceptibility to social proof cues.

**Claude-3.5 Leads on Alignment Biases.** Claude-3.5 achieves the best performance on position (0.832), authority (0.865), distraction (0.878), and diversity (0.914) biases, as well as the highest CoT accuracy (0.745), suggesting stronger constitutional-style constraints (Bai et al., 2022) against demographic and social influence biases.

### 5.2 Impact of the Number of Candidates on Position Bias

Figure 4 shows how position bias intensifies as $n$ increases. For all six judges, robustness rates drop sharply below 0.5 when evaluating 3–4 options, highlighting a critical limitation in multi-answer ranking.

### 5.3 Overall Bias Profile

Figure 5 shows the overall robustness profiles across all bias types and models. The gap between RR and the CR dashed line provides a principled measure of bias magnitude. Sentiment bias consistently produces the largest RR–CR gaps, while verbosity and fallacy biases produce the smallest gaps in frontier models.

### 5.4 Self-Enhancement and Refinement-Aware Biases

Table 5 reports average scores and error rates for self-enhancement and refinement-aware biases. Figure 6 provides a visual comparison.

Table 5: Average score and error rate of self-enhancement bias and refinement-aware bias.

| **Model** | **Sel. Score**$_\downarrow$ | | | **Ref. Score**$_\downarrow$ | | |
|---|---|---|---|---|---|---|
| | Self | Other | Error | Ref | +History | Error |
| **ChatGPT** | 5.21 | 5.72 | 8.91 | 5.23 | 4.94 | 5.80 |
| **GPT-4-Turbo** | 6.98 | 6.90 | **1.16** | 8.31 | 8.45 | 1.66 |
| **GPT-4o** | 7.01 | 6.89 | 1.74 | 7.44 | 7.20 | 3.33 |
| **GLM-4** | 7.73 | 7.64 | 1.18 | 7.64 | 7.73 | **1.15** |
| **Claude-3.5** | 7.04 | 6.55 | 7.48 | 7.51 | 7.68 | 2.17 |
| **Qwen2** | 7.64 | 6.58 | 16.1 | 7.29 | 7.39 | 1.33 |

**Self-Enhancement is Highly Model-Specific.** Self-enhancement error rates vary dramatically: Qwen2 exhibits the highest error rate ($\text{ErrorRate}_{\text{SE}} = 16.1$), indicating a strong tendency to favor its own outputs. Claude-3.5 also shows substantial self-preference (7.48), while ChatGPT shows moderate bias (8.91). In contrast, GPT-4-Turbo (1.16) and GLM-4 (1.18) are nearly unbiased, assigning similar scores to

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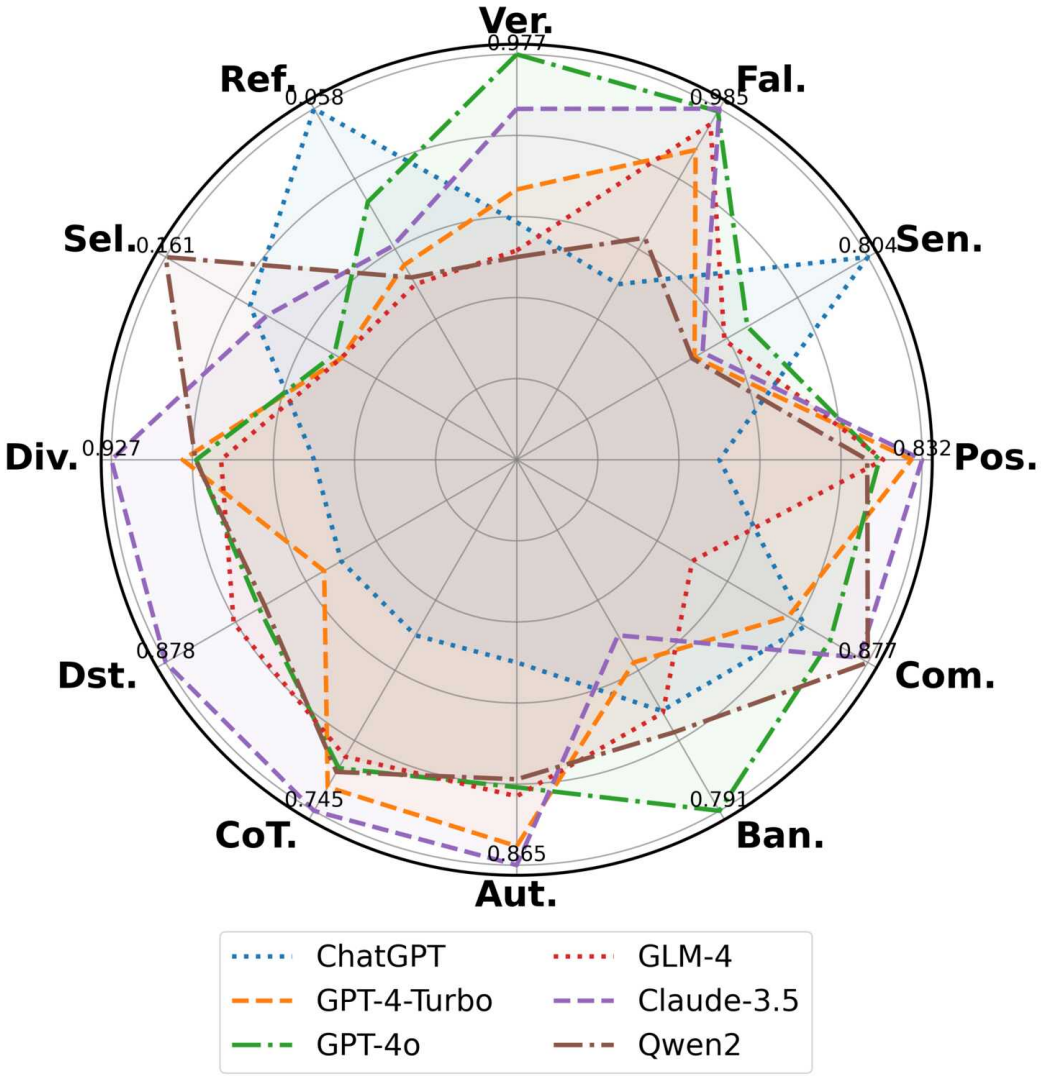


Figure 3: Comparison of robustness rates of all models across bias types. A higher score indicates greater resistance to the corresponding bias. The dashed line represents the consistency rate (CR) for each model.

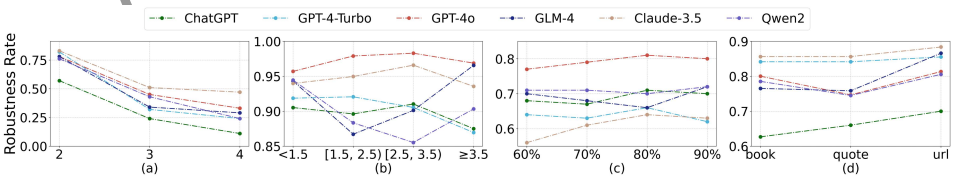


Figure 4: (a) Impact of the number of answers $n$ on the robustness rate in position bias. As the number of candidate responses increases from 2 to 4, robustness rates fall sharply across all models, with RR dropping below 0.5 for 3–4 options. (b) Distribution of judgments across positions.

self-generated and externally-generated answers.
Figure 7 provides a Z-score normalized view of self-enhancement patterns across models.

**Refinement-Aware Bias is More Contained.** For refinement-aware bias, error rates are generally lower. GLM-4 achieves the lowest ($\text{ErrorRate}_{\text{RA}} = 1.15$), while ChatGPT exhibits the highest (5.80).

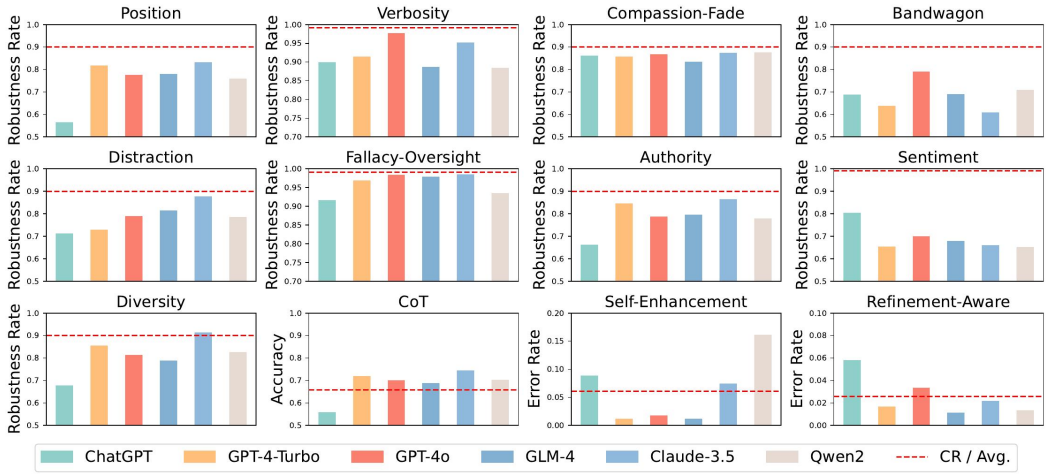


Figure 5: Overall robustness rate for each model and bias type. The dashed line represents the consistency rate (CR), which serves as an upper bound for unbiased behavior. Larger gaps between RR and CR indicate stronger susceptibility to the corresponding bias.

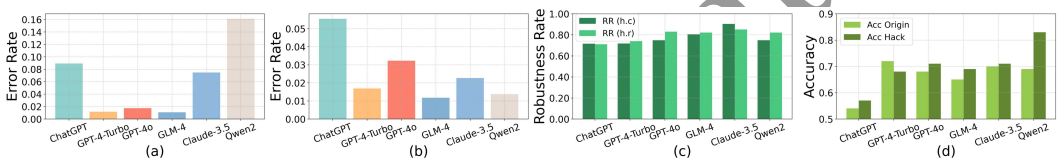


Figure 6: (a) Model error rates for refinement-aware bias ($\text{Err.}_{\text{RA}}$) and (b) self-enhancement bias ($\text{Err.}_{\text{SE}}$). Lower error rates indicate less susceptibility to the corresponding bal.

### 5.5 Chain-of-Thought Bias

CoT prompting improves judge accuracy variably. Claude-3.5 achieves the highest CoT accuracy (0.745) and GLM-4 shows a notable +7pp improvement (0.688). GPT-4-Turbo improves modestly (0.720), while ChatGPT shows the lowest CoT accuracy (0.560).

### 5.6 Bias Recognition Capability

Table 6 presents bias recognition performance for GPT-4-Turbo and Claude-3.5. Claude-3.5 achieves a 0% None Rate (NR) across all bias types, though its Success Rate (SR) varies (e.g., 0.46 for fallacy-oversight). GPT-4-Turbo achieves perfect SR on bandwagon and distraction but struggles with compassion-fade (SR $= 0.48$) and diversity (SR $= 0.46$).

Table 6: Bias recognition performance across different bias types. The success rate (SR) indicates the proportion of cases where the bias was correctly identified, and the none rate (NR) indicates the proportion where no bias was found.

| Bias Type | GPT-4-Turbo | | Claude-3.5 | |
|---|---|---|---|---|
| | SR↑ | NR↓ | SR↑ | NR↓ |
| **Authority** | 0.84 | 0.14 | 0.84 | 0.00 |
| **Bandwagon-effect** | 1.00 | 0.00 | 0.92 | 0.00 |
| **Compassion-fade** | 0.48 | 0.34 | 0.96 | 0.00 |
| **Distraction** | 1.00 | 0.00 | 1.00 | 0.00 |
| **Diversity** | 0.46 | 0.02 | 0.96 | 0.00 |
| **Fallacy-oversight** | 0.52 | 0.04 | 0.46 | 0.00 |
| **Sentiment** | 0.96 | 0.04 | 0.72 | 0.00 |
| **Verbosity** | 0.90 | 0.10 | 1.00 | 0.00 |

## 6 Discussion

**Why Does Sentiment Bias Persist?** Sentiment bias is the most universal failure mode. We hypothesize this arises from RLHF training, where annotators may prefer emotionally engaging responses, causing judges to conflate tone with quality.

**Self-Enhancement and Training Data Homogeneity.** The dramatic variation in self-enhancement (1.16 for GPT-

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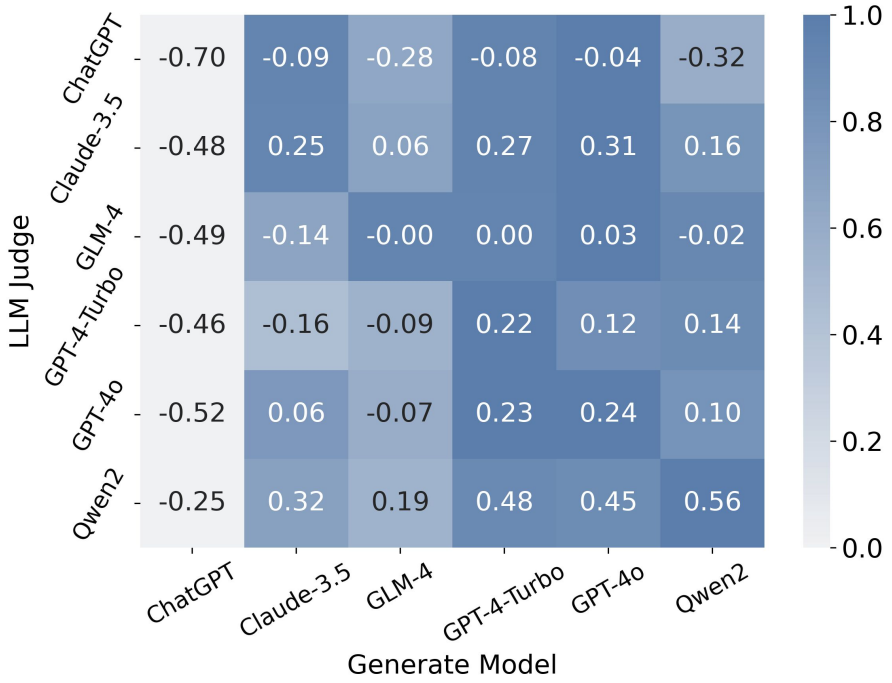


Figure 7: Heat map of Z-score normalized self-enhancement bias scores across models and question categories. Darker cells indicate stronger self-preference relative to other models on that category.

4-Turbo vs. 16.1 for Qwen2) suggests self-preference is closely tied to training data composition and overlap with the evaluated model.

**Position Bias in Multi-Candidate Settings.** Position bias becomes catastrophic (RR $< 0.5$) for $n \geq 3$ candidates—a practically significant finding for real-world multi-response ranking. We recommend explicit position randomization and aggregation across orderings.

**Implications for Reward Modeling.** Biases propagate into DPO (Rafailov et al., 2024) and RLHF (Ouyang et al., 2022) training signals; practitioners should apply position randomization, sentiment neutralization, and multi-model ensembles to mitigate bias-induced noise.

**Limitations.** CALM focuses on text-only English evaluation; multimodal (Chen et al., 2024a) and multilingual (Bang et al., 2023) extensions are important future directions.

## 7 CONCLUSION

We present CALM, a comprehensive automated benchmark for quantifying biases in LLM-as-a-Judge. By framing bias assessment as an attack-and-detect problem with semantics-preserving perturbations across 12 bias types, CALM enables reproducible, scalable evaluation without human annotation. Our evaluation of six state-of-the-art LLMs reveals that (1) sentiment bias is universally the most challenging, (2) self-enhancement bias is highly model-specific, (3) position bias becomes catastrophic in multi-candidate settings, and (4) chain-of-thought prompting provides modest but consistent accuracy improvements.

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